

Baris Guzel

Senior C#/.NET Software Engineer | WPF/MVVM | Real-Time Systems

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PROFESSIONAL SUMMARY

Senior C#/.NET Software Engineer with end-to-end ownership of customer-facing industrial desktop applications, spanning architecture, UI development, deployment, support and continuous improvement. Built and modernised real-time machine-health and CAD/CAM products using C#, .NET Framework, WPF/MVVM, C++/CLI and multithreading, increasing rendering performance from 5 to 100 FPS. Trusted as the primary technical contact for major manufacturers and research organisations, while independently delivering complex CAD/CAM integrations, defining technical roadmaps and mentoring interns.

TECHNICAL SKILLS

Languages: C#, C++, C++/CLI, Python, MATLAB, VB.NET

.NET & Desktop: .NET Framework, WPF, XAML, MVVM, WinForms, multithreading

Real-Time & Industrial: Real-time data acquisition, NI-DAQ, FFT/FRF, signal processing, UDP/TCP, hardware-software integration

CAD/CAM & Interoperability: Mastercam SDK, Siemens NX, managed/unmanaged interoperability, XML ribbon customisation

Web, Cloud & Data: JavaScript, HTML, CSS, Node.js, AWS Lambda, Amazon S3, Amazon CloudWatch, MySQL, SQLite

Engineering & Delivery: Software architecture, API/SDK design, performance optimisation, technical roadmapping, Git, Bitbucket, Jira, Confluence, Agile/Scrum, Advanced Installer

Development Tools: Visual Studio, Visual Studio Code, Figma

PROFESSIONAL EXPERIENCE

Productive Machines Limited

Full Stack Developer (C#/.NET & Industrial Software) | Rotherham, UK | Jul 2023 - Present

- Own end-to-end technical delivery of software products contributing over 30% of company revenue, spanning architecture, implementation, deployment, customer support and continuous improvement.
- Designed and developed Tap Starter, a real-time C#/.NET Framework machine-health analytics desktop application integrating NI-DAQ hardware for IEPE impact-hammer and accelerometer data acquisition, FFT/FRF analysis and industrial tap-testing workflows.
- Led the migration from WinForms to WPF using MVVM and XAML, introducing GPU-accelerated rendering and increasing visualisation performance from approximately 5 to 100 FPS (20x), improving responsiveness and usability.
- Serve as the primary technical contact for major manufacturing and research customers, translating feedback, defect reports and enhancement requests into prioritised roadmaps and production releases.
- Built AWS-connected in-app news and maintenance notifications alongside an automatic update workflow, enabling customers to receive service announcements and software releases directly within the product.
- Independently architected and developed a Mastercam add-on using C#, C++/CLI, WPF, XAML and the Mastercam SDK to extract tooling and operation data, integrate server-based and on-premises optimisation services, and re-import optimised toolpaths.
- Resolved a critical Mastercam integration blocker after external escalation using controlled multithreading, thread-priority management, SEH and OLE message filtering; also developed responsive, non-blocking Siemens NX optimisation workflows.
- Built a reusable company-wide SDK for MySQL access on AWS; contribute to JavaScript and Node.js web development, mentor interns and participate in the Scrum Master rotation.

Rapsodo PTE Limited

Software Engineer (Remote) | Singapore | Feb 2023 - Jul 2023

- Contributed to C++ Wi-Fi driver functionality and implemented multithreading for SkyTrak and SkyTrak+ launch monitors, improving real-time shot-data capture stability.
- Contributed to UDP/TCP protocol design for SDK-to-firmware communication and integrated AES-256 encryption, reducing live-tracking latency and securing wireless data transmission.
- Contributed to Unity and C# optimisation for real-time shot visualisation and application responsiveness.

TÜBİTAK BİLGEM İLTAREN

Candidate Researcher | Ankara, Turkey | Aug 2022 - Feb 2023

- Developed a C# XML parser to process structured data for internal data-integration workflows.
- Applied MATLAB image-processing techniques, including FFT, upsampling and downsampling, to improve real-time data visualisation.
- Optimised a real-time C++ streaming pipeline to reduce packet loss and improve system responsiveness.

Arçelik / Beko

Software Engineer Intern | Eskişehir, Turkey | Jul 2022 - Aug 2022

- Designed and developed a Digital Twin SCADA system using Node-RED, Node.js and Siemens PLCs, enabling engineers to remotely locate faults across a large manufacturing facility.
- Created real-time system visualisations in Inkscape and backend monitoring and control workflows, accelerating fault diagnosis and helping minimise production-line downtime.

KoçSistem

Online Support Specialist (Remote) | İstanbul, Turkey | Oct 2021 - Jul 2022

- Managed operational IT support within Operational Planning and Integrated IT Management, raising and tracking ITIL change records while monitoring SLAs, service requests, risks and impacts.
- Developed RPA automation scripts to streamline internal workflows, reducing processing time and operational overhead.

Technology Planning Intern (Remote) | İstanbul, Turkey | Aug 2021 - Sep 2021

- Supported the Technology Planning Department by reviewing domain management, networking and Agile workflows, collaborating across teams to address process bottlenecks and producing technical documentation, process diagrams and client-facing presentations.

EDUCATION

Middle East Technical University (METU)

BSc Electrical and Electronics Engineering | Ankara, Turkey | 2018 - 2023

SELECTED TECHNICAL PROJECTS

CopyCat Robotic System (Final-Year Project) | C++, Qt, OpenCV, Raspberry Pi

- Led a five-person team to build a two-vehicle Master-Agent robotic system using camera tracking and sensor fusion for real-time path following and constant-distance control; earned the Good Product Design Award as the only team to complete the project.

FPGA Pipelined RISC Processor | Verilog, ModelSim, Terasic DE1-SoC

- Designed and implemented a pipelined RISC processor with hazard detection and branch prediction, reducing pipeline stalls and improving instruction throughput; verified through ModelSim simulation and on-board FPGA testing.

ADDITIONAL INFORMATION

Languages: English, Turkish and Kurdish - fluent | **Volunteering:** Community event organiser